

Code Decode Lab - Hands-On Python Tutorial

Lab Objective

In this lab, you'll build a utility that can encode and decode multilingual strings into bytes and back. This hands-on activity will help you understand how Python handles text vs. binary data, especially useful when working with files or transferring data across systems.

Key Concepts

- Strings are human-readable text (Unicode).
- Bytes are how text is stored/transmitted in memory/files.
- UTF-8 is a standard encoding that works with many languages.
- Encoding: string -> bytes | Decoding: bytes -> string

Part 1: Multilingual Word List

```
words = [  
    'Cafe',    # French  
    'Privet',  # Russian  
    'Konnichiwa', # Japanese  
    'Marhaban', # Arabic  
    'Shalom',   # Hebrew  
    'Namaste',  # Hindi  
    'Annyeonghaseyo' # Korean  
]
```

Part 2: Encoding & Decoding Functions

```
def encode_word(word, encoding="utf-8"):  
    return word.encode(encoding)  
  
def decode_word(byte_data, encoding="utf-8", errors="strict"):  
    return byte_data.decode(encoding, errors=errors)
```

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Part 3: Code Decode Menu

Create a simple command-line interface:

1. Encode a word to bytes
2. Decode a byte string

Accept user input for encoding type and error handling.

Use ``eval()`` to safely parse byte string inputs.

Part 4: Challenges

Challenge 1: Allow different encodings (already built into the menu).

Challenge 2: Add error handling using 'ignore' and 'replace'.

Challenge 3: Save and load encoded data to/from a file.

Challenge 4: Batch encode/decode a list of words and print results.

Challenge 3 Code Snippet

```
def save_encoded_to_file(word, filename, encoding="utf-8"):
    with open(filename, "wb") as f:
        f.write(encode_word(word, encoding))

def load_and_decode_from_file(filename, encoding="utf-8"):
    with open(filename, "rb") as f:
        return decode_word(f.read(), encoding)
```

Challenge 4 Code Snippet

```
def batch_encode_decode(word_list, encoding="utf-8"):
    for word in word_list:
        encoded = encode_word(word, encoding)
        decoded = decode_word(encoded, encoding)
        print(f'{word} -> {encoded} -> {decoded}')
```